
Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Eguny 2, Robert
Date of birth:	5/8/1991
Subject:	080- Principles of flight
Result:	0 %
Time used:	00:00:15 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	223	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
2	210	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
3	185	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
4	194	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
5	198	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
6	213	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
7	205	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
8	189	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
9	58	81.1.1.1 Laws and definitions		0,00	1,00
10	82	81.1.1.1 Laws and definitions		0,00	1,00
11	48	81.1.1.1 Laws and definitions		0,00	1,00
12	23	81.1.2.4 Centre of pressure and aerodynamic centre		0,00	1,00
13	90	81.1.4.2 Induced drag		0,00	1,00
14	74	81.1.4.2 Induced drag		0,00	1,00
15	38	81.1.8.2 The stall speed		0,00	1,00
16	28	81.1.8.5 Special phenomena of stall		0,00	1,00
17	91	81.2.1.3 Influence of temp and alt on M		0,00	1,00
18	16	81.2.2.2 Oblique shock waves		0,00	1,00
19	66	81.2.2.2 Oblique shock waves		0,00	1,00
20	9	81.2.2.2 Oblique shock waves		0,00	1,00
21	42	81.2.3.4 Effect on pitching moment		0,00	1,00
22	104	81.4 STABILITY		0,00	1,00
23	86	81.4.1.1 Basics and definitions		0,00	1,00
24	87	81.4.3.3 Neutral point		0,00	1,00
25	21	81.5.2.4 Location of centre of gravity		0,00	1,00
26	77	81.5.8.3 Stabiliser trim		0,00	1,00
27	10	81.5.8.3 Stabiliser trim		0,00	1,00
28	32	81.5.8.3 Stabiliser trim		0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	96	81.6.1.1	Flutter	0,00	1,00
30	108	81.7	PROPELLERS	0,00	1,00
31	107	81.7	PROPELLERS	0,00	1,00
32	50	81.7.1.3	Fixed and variable pitch/constant speed	0,00	1,00
33	73	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
34	64	81.7.4.1	Torque reaction	0,00	1,00
35	83	81.7.4.2	Gyroscopic precession	0,00	1,00
36	8	81.7.4.2	Gyroscopic precession	0,00	1,00
37	24	81.8.1.2	Straight steady climb	0,00	1,00
38	69	81.8.1.2	Straight steady climb	0,00	1,00
39	29	81.8.1.4	Straight steady glide	0,00	1,00
40	68	81.8.1.5	Steady coordinated turn	0,00	1,00
Total:0%				0,00	40,00

Attachment 2: List of result by topic

Licence

Date: 28.07.2014

Examinee: Eguny 2, Robert

Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	0,00	40,00	0,00
81.1. SUBSONIC AERODYNAMICS	8	0,00	8,00	0,00
81.2. HIGH SPEED AERODYNAMICS	5	0,00	5,00	0,00
81.4. STABILITY	3	0,00	3,00	0,00
81.5. CONTROL	4	0,00	4,00	0,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	7	0,00	7,00	0,00
81.8. FLIGHT MECHANICS	4	0,00	4,00	0,00
Total	40	0,00	40,00	0,00

1

Which is the correct symbol for the stalling speed or the minimum steady flight speed at which the airplane is controllable?

☒ $V(S)$.

☐ $V(S1)$.

☐ $V(S0)$.

2

A rule of thumb for flying a final approach is to maintain a speed that is

- ☒ 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☐ twice the glider's stall speed plus half the estimated windspeed.
- ☐ 50 percent above the glider's stall speed, regardless of windspeed.

3

If the landing gear on an airplane moves forward during retraction, the total moment will

☐ increase.

☐ remain the same.

☒ decrease.

4

If the CG of an aircraft is moved from the aft limit to beyond the forward limit, how will it affect the cruising and stalling speed?

- ☐ Increase both the cruising speed and stalling speed.
- ☒ Decrease the cruising speed and increase the stalling speed.
- ☐ Decrease both the cruising speed and stalling speed.

5

A pilot of a high-performance airplane should be aware that flying a steeper-than-normal VASI glide slope angle may result in

- ☒ increased landing rollout.
- ☐ landing short of the runway threshold.
- ☐ a hard landing.

6

Which is true relating to the direction in which turns should be made during slope soaring?

- ☐ All turns should be made downwind toward the slope.
- ☒ All reversing turns should be made into the wind away from the slope.
- ☐ All reversing turns should be made to the left.

7

What could result if a glider pilot releases while in the low-tow position during an aerotow?

- ☐ Glider may be forced into the towplane's wake turbulence.
- ☒ Tow ring may strike and damage the glider after release.
- ☐ Nose of the glider would tend to pitch up after release.

8

When one engine fails on a twin-engine airplane, the resulting performance loss

- ☐ is approximately 50 percent since 50 percent of the normally available thrust is lost.
- ☐ reduces cruise indicated airspeed by 50 percent or more.
- ☒ may reduce the rate of climb by 80 percent or more.

9

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

- I. The dynamic pressure in the throat is lower than in the undisturbed airflow.
II. The total pressure in the throat is the same as in the undisturbed airflow.

- ☐ I is correct, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is incorrect, II is correct.

10

The SI units of air density (I) and force (II) are..... &... respectively:

☒ (I) kg / m^3 , (II) N.

☐ (I) N / m^3 , (II) N.

☐ (I) N / kg , (II) kg.

☐ (I) kg / m^2 , (II) kg.

11

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

- I. The static pressure in the undisturbed airflow is lower than in the throat.
II. The speed in the undisturbed airflow is lower than in the throat.

- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is correct.

12

Assuming no flow separation and no compressibility effects the location of the centre of pressure of a symmetrical aerofoil section:

- ☒ is at approximately 25% chord irrespective of angle of attack.
- ☐ is at approximately 50% chord irrespective of angle of attack.
- ☐ moves backward when the angle of attack decreases.
- ☐ moves forward when the angle of attack decreases.

13

Excluding constants, the coefficient of induced drag (CDi) is the ratio of:

- ☐ CL and CD.
- ☐ CL^2 and S (wing surface).
- ☒ CL^2 and AR (aspect ratio).
- ☐ CL and b (wing span).

14

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☐ total drag increases by a factor of 4.
- ☒ induced drag increases by a factor of 4.
- ☐ lift increases by a factor of 4.
- ☐ angle of attack increases by a factor of 1/4.

15

The stall speed increases, when: (all other factors of importance being constant)

- ☐ weight decreases.
- ☐ spoilers are retracted.
- ☐ minor altitude changes occur e.g. 0-10.000 ft.
- ☒ pulling out of a dive.

16

A stick pusher:

- ☐ pushes the elevator control forward prior to stick shaker activation.
- ☐ pushes the elevator control to avoid a stall at a negative load factor.
- ☐ vibrates the elevator control.
- ☒ pushes the elevator control forward when a specified value of angle of attack is exceeded.

17

If the altitude is increased and the TAS remains constant in the troposphere under standard atmospheric conditions, the Mach number will:

- ☐ not change.
- ☐ decrease.
- ☒ increase.
- ☐ increase or decrease, depending on the type of aeroplane.

18

Compared with an oblique shock wave a normal shock wave has a:

- ☐ higher total temperature.
- ☐ higher total pressure.
- ☒ higher loss in total pressure.
- ☐ lower static temperature.

19

Which of these statements about an oblique shock wave are correct or incorrect?

- I. The static temperature in front of an oblique shock wave is higher than behind it.
II. The static pressure in front of an oblique shock wave is lower than behind it.

- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

20

Which of these statements about an oblique shock wave are correct or incorrect?

- I. The local speed of sound in front of an oblique shock wave is higher than behind it.
II. The Mach number in front of an oblique shock wave is higher than behind it.

- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

21

What is the effect of exceeding critical mach. No. on the stick force stability of an aeroplane with swept-back wings without any form of stability augmentation?

- ☐ An increase, due to shock wave formation in the wing root area.
- ☐ No effect, because critical mach. No. is not relevant when considering stick force stability.
- ☒ A decrease, due to loss of lift in the wing root area.
- ☐ No effect, because stick force stability is independent of Mach number.

22

If an aircraft has negative dynamic and positive static stability, this will result in

- ☐ convergent oscillations.
- ☒ divergent oscillations.
- ☐ undamped oscillations.

23

Positive static stability of an aeroplane means that following a disturbance from the equilibrium condition:

- ☐ the tendency is to move with an oscillatory motion of decreasing amplitude.
- ☐ the initial tendency is to diverge further from its equilibrium condition.
- ☒ the initial tendency is to return towards its equilibrium condition.
- ☐ the tendency is to move with an oscillatory motion of increasing amplitude.

24

Which of the following statements about static longitudinal stability is correct?

- I. A requirement for positive static longitudinal stability of an aeroplane is, that the neutral point is behind the centre of gravity.
- II. A wing with positive camber provides a positive contribution to static longitudinal stability, when the centre of gravity of the aeroplane is in front of the aerodynamic centre of the wing.

- ☒ I is correct, II is correct.
- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.

25

Which statement is about CG limits is correct?

- ☐ If the aft CG limit is correctly chosen, the forward CG limit is automatically determined as well.
- ☐ The forward CG limit is determined by stability considerations only.
- ☒ The forward CG limit is mainly determined by the amount of pitch control available from the elevator.
- ☐ The aft CG limit is determined by the maximum elevator deflection available.

26

The most important factor determining the required position of the Trimmable Horizontal Stabiliser (THS) for take off is the:

- ☒ **position of the aeroplane's centre of gravity.**
- ☐ **centre of gravity position of the fuel.**
- ☐ **stall speed.**
- ☐ **total mass of the aeroplane.**

27

Comparing the differences between a horizontal trimmable stabiliser and an elevator trim tab, which statement is correct?

- I. A stabiliser trim is less suitable for jet transport aeroplanes because of their large speed range.
II. A trim tab is a more powerful means of trimming.

☒ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

28

When comparing a stabiliser trim system with an elevator trim system, which of these statements is correct?

- ☒ a stabiliser trim is able to compensate larger changes in pitching moments.
- ☐ an elevator trim is more suitable for aeroplanes with a large CG range.
- ☐ an elevator trim is able to compensate larger changes in pitching moments.
- ☐ a stabiliser trim is more sensitive to flutter.

29

Wing flutter may be caused by a:

- ☐ combination roll control reversal and low speed stall.
- ☐ combination of fuselage bending and wing torsion.
- ☒ combination of bending and torsion of the wing structure.
- ☐ aerodynamic wing stall at high speed.

30

Blade angle of a propeller is defined as the angle between the

- ☐ angle of attack and chord line.
- ☐ angle of attack and line of thrust.
- ☒ chord line and plane of rotation.

31

The distance a propeller actually advances in one revolution is

☒ effective pitch.

☐ geometric pitch.

☐ twisting.

32

The angle of attack of a fixed pitch propeller blade increases when:

- ☐ velocity and RPM increase.
- ☐ velocity and RPM decrease.
- ☒ RPM increases and forward velocity decreases.
- ☐ forward velocity increases and RPM decreases.

33

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

- I. A constant speed propeller reduces fuel consumption over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.
- ☒ I is correct, II is incorrect

34

The torque reaction of a rotating fixed pitch propeller will be greatest at:

- ☐ low aeroplane speed and low engine power.
- ☒ low aeroplane speed and maximum engine power.
- ☐ high aeroplane speed and low engine power.
- ☐ high aeroplane speed and maximum engine power.

35

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch down produces left yaw.
II. Left yaw produces pitch down.

- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☒ I is correct, II is incorrect.

36

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch up produces left yaw.
II. Right yaw produces pitch up.

- ☐ I is correct, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.
- ☒ I is incorrect, II is incorrect.

37

An aeroplane climbs to cruising level with a constant pitch attitude and maximum climb thrust, (assume no supercharger).
How do the following variables change during the climb? (Gamma = flight path angle)

- ☐ Gamma decreases, angle of attack increases, IAS remains constant.
- ☐ Gamma remains constant, angle of attack remains constant, IAS decreases.
- ☐ Gamma decreases, angle of attack remains constant, IAS decreases.
- ☒ Gamma decreases, angle of attack increases, IAS decreases.

38

Given:
Aeroplane mass: 50 000kg.
Lift/Drag ratio: 12.
Thrust per engine: 28 000N.
Assumed g: 10m/s².

For a straight, steady, wings level climb of a three-engine aeroplane, the one-engine inoperative climb gradient is:

☐ 8.5%.

☒ 2.9%.

☐ 8.0%.

☐ 9.7%.

39

Which of the following increases the maximum duration of a glide?

- ☐ A tailwind.
- ☒ A decrease in mass.
- ☐ A headwind.
- ☐ An increase in mass.

40

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a higher mass:

- ☐ will turn with a larger turn radius.
- ☐ will have a higher rate of turn.
- ☐ will turn with a smaller turn radius.
- ☒ will turn with the same radius, but might stall.